

Type-B Self-Similarity of the Fake-Degree Sum at a Root of Unity: A Discovery Report

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Abstract

Let $W_B = \mathbb{Z}/2 \wr S_n$ be the hyperoctahedral group of rank n , with degrees $\{2, 4, \dots, 2n\}$. For $\zeta_e = e^{2\pi i/e}$, define the type-B fake-degree sum

$$q_e^{B(n)} := \sum_{(\alpha, \beta)} \frac{\Psi_{(\alpha, \beta)}^B(\zeta_e)}{z_{(\alpha, \beta)}^B} p_\alpha(x) p_\beta(y),$$

where the sum is over signed cycle types (α, β) with $|\alpha| + |\beta| = n$, $\Psi_{(\alpha, \beta)}^B(t) := \prod_{i=1}^n (1 - t^{2i}) / [\prod_{a \in \alpha} (1 - t^a) \prod_{b \in \beta} (1 + t^b)]$, and $z_{(\alpha, \beta)}^B := 2^{\ell(\alpha) + \ell(\beta)} z_\alpha z_\beta$.

Numerical experiment across 24 test pairs (n, e) — with exact arithmetic in $\mathbb{Z}[\zeta_e]$ — gives *exact zero residual* for the self-similarity identity

$$q_e^{B(n)} = (p_e^B)^{\lfloor n/e \rfloor} \cdot q_e^{B(n \bmod e)}$$

where p_e^B depends on the parity of e :

$$p_e^B = \begin{cases} p_{(e)}(x) & e \text{ odd} \\ p_{(e/2, e/2)}(y) & e \text{ even.} \end{cases}$$

Both cases have coefficient exactly 1 in the bipartition power-sum basis.

The odd- e statement is a mechanical translation of the type-A theorem [?] with the negative-cycle bipartition β as an inert passenger; the proof is outlined here and marked for a dedicated **PROVE** session. The even- e statement is genuinely new: it requires a strictly stronger *negative-side* support theorem (killing bipartitions with too few $(e/2)$ -cycles) together with a companion cyclotomic identity $\Psi_{\emptyset, (e/2, e/2)}^B(\zeta_e) = 2e^2$. Both ingredients look clean from the compute.

1 Setup

Let $W_B = \mathbb{Z}/2 \wr S_n$, with reflection representation V of dimension n . An element $w \in W_B$ has a *signed cycle type* (α, β) where $\alpha \vdash a$ are the positive cycles and $\beta \vdash b$ the negative cycles, $a + b = n$. The characteristic polynomial of w acting on V is

$$\det(1 - tw|_V) = \prod_{a \in \alpha} (1 - t^a) \prod_{b \in \beta} (1 + t^b).$$

The Chevalley–Molien series for the coinvariant algebra of W_B gives

$$\Psi_{(\alpha, \beta)}^B(t) := \frac{\prod_{i=1}^n (1 - t^{2i})}{\prod_{a \in \alpha} (1 - t^a) \prod_{b \in \beta} (1 + t^b)} = \sum_{(\lambda, \mu)} \tilde{f}_{(\lambda, \mu)}^B(t) \chi^{(\lambda, \mu)}((\alpha, \beta)), \quad (1)$$

where $\tilde{f}_{(\lambda,\mu)}^B(t)$ is the type-B fake degree of the irreducible W_B -representation indexed by the bipartition $(\lambda, \mu) \vdash n$, and $\chi^{(\lambda,\mu)}$ its character. The right-hand side is a polynomial in $\mathbb{Z}[t]$.

The type-B centraliser order is $|Z_{W_B}(w)| = z_{(\alpha,\beta)}^B := 2^{\ell(\alpha)+\ell(\beta)} z_\alpha z_\beta$, and Frobenius reciprocity gives

$$q_e^{B(n)} := \sum_{|\alpha|+|\beta|=n} \frac{\Psi_{(\alpha,\beta)}^B(\zeta_e)}{z_{(\alpha,\beta)}^B} p_\alpha(x) p_\beta(y). \quad (2)$$

This is the type-B analogue of the type-A object $q_e^{(n)} = \sum_\lambda \tilde{f}_\lambda(\zeta_e) s_\lambda$ from [?], written in the power-sum basis of W_B -class functions.

2 Main Empirical Statement

Theorem 1 (Type-B Self-Similarity — Empirical). *For every (n, e) with $n \geq 0$, $e \geq 2$, and $k := \lfloor n/e \rfloor$, $r_0 := n \bmod e$,*

$$q_e^{B(n)} = (p_e^B)^k \cdot q_e^{B(r_0)}$$

in the bipartition power-sum algebra $\mathbb{Z}[\zeta_e][p_1(x), p_2(x), \dots; p_1(y), p_2(y), \dots]$, where

$$p_e^B = \begin{cases} p_{(e)}(x) & e \text{ odd}, \\ p_{(e/2, e/2)}(y) & e \text{ even}. \end{cases}$$

Verified with exact arithmetic in $\mathbb{Z}[\zeta_e]$ (cyclotomic reduction, no floating point) at 24 test pairs

$$(n, e) \in \{(3, 2), (4, 2), (5, 2), (7, 2), (9, 2); (5, 3), (6, 3), (7, 3), (8, 3), (9, 3), (11, 3);$$

$$(5, 4), (7, 4), (8, 4), (9, 4), (11, 4); (7, 5), (9, 5), (11, 5), (13, 5); (6, 6), (7, 6), (8, 6), (9, 6)\}.$$

Residuals are exactly zero on every test pair. Support sizes range from 1 to 20 nonzero coefficients (largest at $(n, e) = (9, 5)$).

Verification code: `~/projects/probes/2026-08-08-type-b-self-similarity/probe.py`.

Corollary 2 (Parity dichotomy). *The factor p_e^B is supported entirely on the positive alphabet x when e is odd, and entirely on the negative alphabet y when e is even. In particular, the restriction $p_e^B|_{y=0}$ recovers the type-A p_e when e is odd but is identically zero when e is even.*

Observation 3 (Two-sided support). Numerically, the support of $q_e^{B(r_0)}$ obeys:

- *Odd e .* All bipartitions $(\alpha', \beta) \vdash r_0$ appear with generically nonzero coefficient. Support sizes match $\#\{(\alpha', \beta) : |\alpha'| + |\beta| = r_0\}$ exactly.
- *Even e .* Only a small strict subset of bipartitions survive. Examples:
 - $(r_0, e) = (3, 4)$: only 2 of 10 bipartitions of 3 survive, namely $(\emptyset, (2, 1))$ and $((1), (2))$.
 - $(r_0, e) = (3, 6)$: only 1 of 10 survives, namely $(\emptyset, (3))$.

The even- e pattern is a genuinely new phenomenon — a strictly stronger support theorem that constrains not just the positive side (as in type A) but also the negative side.

3 Sketch of the Odd- e Proof

The odd- e case is a mechanical translation of the type-A theorem [?], with the negative-cycle partition β carried through as an inert passenger. We outline the steps here; the full proof is deferred to a dedicated PROVE session (see `state/PROVE.md`).

Step 1: Numerator cancellation

For odd e , $\zeta_e^{2i} = 1 \iff e \mid 2i \iff e \mid i$ (since $\gcd(2, e) = 1$). Furthermore for each i not a multiple of e , the residue $2i \bmod e$ is a permutation of $i \bmod e$ across $\{1, \dots, e-1\}$ (again since 2 is invertible mod e). Hence

$$\prod_{i=1}^n (1 - t^{2i}) \Big|_{t=\zeta_e} = \prod_{i: e \nmid i, 1 \leq i \leq n} (1 - \zeta_e^{2i}) = \prod_{i: e \nmid i, 1 \leq i \leq n} (1 - \zeta_e^i).$$

This is the same numerator as the type-A polynomial $\prod_{i=1}^n (1 - t^i)$ evaluated at ζ_e . Also, the numerator has exactly k formal zeros (from $i \in \{e, 2e, \dots, ke\}$), matching the type-A count.

Step 2: Negative-cycle denominator is a nonzero constant

For odd e , no integer b satisfies $\zeta_e^b = -1$, so $\prod_{b \in \beta} (1 + \zeta_e^b)$ is a nonzero element of $\mathbb{Z}[\zeta_e]$ for any β . In particular this factor never contributes a pole or zero to $\Psi_{(\alpha, \beta)}^B(\zeta_e)$; it enters the coefficient as a benign multiplicative constant.

Step 3: Support theorem (odd e)

Combining Steps 1–2 with the pole-order count from α , the type-A support argument [?] translates *verbatim* to give:

$$\frac{\Psi_{(\alpha, \beta)}^B(\zeta_e)}{z_{(\alpha, \beta)}^B} \neq 0 \iff \alpha = (e^k, \alpha') \text{ with } \alpha' \vdash r_1, \alpha' \text{ has parts } < e, \beta \vdash r_2, r_1 + r_2 = r_0.$$

Step 4: Byproduct formula and one-line collapse

The type-A byproduct closed form [?, eq. (??)] translates to (odd e):

$$c_{(e^k, \alpha'), \beta}^{B, (n)} = \frac{N_\beta(\zeta_e)}{e^k z_{(e^k, \alpha'), \beta}^B} \prod_{r=1}^{e-1} (1 - \zeta_e^r)^{k+1[r \leq r_0] - m_r(\alpha')}, \quad (3)$$

where $N_\beta(\zeta_e) := \prod_{b \in \beta} (1 + \zeta_e^b)$. The k -dependence factors as $[\prod_r (1 - \zeta_e^r)]^k = e^k$, cancelling the e^k prefactor, exactly as in [?]. The remainder is the same expression at parameter r_0 :

$$c_{(e^k, \alpha'), \beta}^{B, (n)} = c_{\alpha', \beta}^{B, (r_0)}.$$

Since $p_{(e^k, \alpha')}(x) = p_e(x)^k p_{\alpha'}(x)$, we conclude $q_e^{B, (n)} = p_e(x)^k q_e^{B, (r_0)}$, i.e. $p_e^B = p_e(x) = p_{(e)}(x)$ for odd e . $\square$

Sanity check

At $\beta = \emptyset$ the type-B formula collapses to the type-A one: $N_{\emptyset}(\zeta_e) = 1$ and $z_{(e^k, \alpha'), \emptyset}^B = 2^k \cdot z_{(e^k, \alpha')}$. The extra 2^k comes from the type-B centraliser, and does *not* appear in the type-A statement — but note we are computing coefficients in the $p_{\alpha}(x)p_{\beta}(y)$ basis, not the type-A p_{α} basis. Under the projection $y \mapsto 0$ this is consistent.

4 The Even- e Case Is Substantive

For even e , two features change:

Feature 1: The negative-cycle factor $\prod_b(1 + \zeta_e^b)$ has zeros

Namely $1 + \zeta_e^b = 0 \iff \zeta_e^b = -1 \iff b \equiv e/2 \pmod{e}$. Hence a negative b -cycle contributes a zero to the denominator of $\Psi_{(\alpha, \beta)}^B(\zeta_e)$ whenever $e/2 \mid b$ (formally: whenever $b \in \{e/2, 3e/2, 5e/2, \dots\}$).

Feature 2: The numerator has an unexpected extra zero

For even e , the numerator zero $\prod_{i=1}^n(1 - t^{2i})|_{t=\zeta_e} = 0$ arises when $e \mid 2i$, i.e. $e/2 \mid i$. This gives $\lfloor n/(e/2) \rfloor = 2k + 1[r_0 \geq e/2]$ zeros — roughly *twice* as many as in the odd case.

Consequence: even- e support theorem

The pole/zero balance now demands the denominator contribute matching order. Since positive a -cycles with $e \mid a$ give k zeros (as in odd e), the remaining zeros must come from either negative $(e/2)$ -cycles (each contributing one) or negative $(3e/2)$ -cycles, etc. Empirically (Observation ??), the survivors at $r_0 < e$ are exactly those bipartitions where the negative side supplies the “missing” extra zeros. This is a strictly stronger support theorem than type A — the negative side must carry a specific arithmetic.

The even- e factor $p_e^B = p_{(e/2, e/2)}(y)$

At $(\alpha, \beta) = (\emptyset, (e/2, e/2))$ one computes directly:

$$\Psi_{\emptyset, (e/2, e/2)}^B(\zeta_e) = \lim_{t \rightarrow \zeta_e} \frac{\prod_{i=1}^e (1 - t^{2i})}{(1 + t^{e/2})^2} = 2e^2,$$

using L'Hôpital: $(1 + t^{e/2})$ has a double zero, and $\prod_{i=1}^e (1 - t^{2i})$ has a matching double zero at $t = \zeta_e$ from the factors $i = e/2$ and $i = e$. The centraliser is $z_{\emptyset, (e/2, e/2)}^B = 2^2 \cdot z_{(e/2, e/2)} = 4 \cdot ((e/2)^2 \cdot 2!) = 2e^2$. Hence the coefficient equals $\Psi/z^B = (2e^2)/(2e^2) = 1$, matching the extracted p_e^B .

Sketch of the even- e proof strategy

The full even- e proof requires:

1. A support theorem: characterise which bipartitions $(\alpha, \beta) \vdash n$ have $\Psi_{(\alpha, \beta)}^B(\zeta_e)/z_{(\alpha, \beta)}^B \neq 0$. Compute suggests the pattern is:

$$\alpha = (e^k, \alpha'), \quad \beta = ((e/2)^{2k'}, \beta''), \quad |\alpha'| + |\beta''| + e \cdot k + (e/2) \cdot 2k' = n,$$

with α' having parts $< e$ and β'' having parts not in $\{e/2, e, 3e/2, \dots\}$ (needs verification).

2. A companion cyclotomic identity: $\prod_{r=1}^{e-1} (1 - \zeta_e^r) \cdot [\text{negative-side product}] = 2e^2$ or a suitable factorisation.
3. The one-line collapse: the byproduct formula at parameter n has a k -shift on the positive side and a k' -shift on the negative side; each shift is absorbed by its own cyclotomic identity.

The compute strongly suggests all three ingredients are clean; the full proof is a task for a dedicated PROVE session.

5 Consequences and Impact

Corollary 4 (n -independence of the type-B residue factor). *For fixed e , the residue factor $F_e^{B,(n)} := q_e^{B(n)} / (p_e^B)^{\lfloor n/e \rfloor}$ depends on n only through $r_0 = n \bmod e$.*

Corollary 5 (Multiplicative recursion). $q_e^{B(n)} = p_e^B \cdot q_e^{B(n-e)}$ for $n \geq e$.

Corollary 6 (Fundamental domain). *The sequence $(q_e^{B(n)})_{n \geq 0}$ is a p_e^B -lattice over the “basic” values $q_e^{B(0)}, q_e^{B(1)}, \dots, q_e^{B(e-1)}$.*

Implications for the composite- d paper

Combined with the type-A self-similarity [?] and the type-A support theorem [?], Theorem ?? would extend the composite- d story to classical types:

- **Odd e :** the story is essentially identical to type A. Same mechanism, same cyclotomic collapse.
- **Even e :** genuinely new phenomenon. The factor p_e^B “lives on the negative side” and represents pairs of $(e/2)$ -cycles rather than e -cycles. This should have a clean representation-theoretic interpretation — perhaps in terms of W_B ’s regular elements at ζ_e (which for even e are precisely the products of $2k$ commuting reflections of appropriate order).

The parity dichotomy is aesthetically striking: at odd e , type B is just type A carrying an inert cocycle; at even e , the second alphabet becomes the dominant partner. This looks like a shadow of a $\mathbb{Z}/2$ -grading on the underlying Chevalley–Molien mechanism.

Remark 7 (Ayyer–Kumari as the classical-groups analogue of Verschiebung). The 2026-08-06 browse identified Ayyer–Kumari’s odd-power-of-even-root character factorisation [?] as the structural type-B/C/D analogue of the Verschiebung operator that the 2026-08-05 dream projected as the “right” tool. As with the two type-A theorems this week (support and self-similarity), the Ayyer–Kumari machinery may not be directly needed: the compute suggests Chevalley–Molien plus the classical cyclotomic identities suffices. Verschiebung was projected twice, needed zero times in type A; the pattern may repeat here. This is a hypothesis to be tested by the even- e proof.

6 Test-Pair Table

(n, e)	k	r_0	p_e^B shape	support of $q_e^{B(n)}$	residual
(3, 2)	1	1	$(\emptyset, (1, 1))$	2 (of 5)	0 exact
(4, 2)	2	0	$(\emptyset, (1, 1))$	1 (of 10)	0 exact
(5, 3)	1	2	$((3), \emptyset)$	5 (of 10)	0 exact
(6, 3)	2	0	$((3), \emptyset)$	1 (of 10)	0 exact
(7, 3)	2	1	$((3), \emptyset)$	2 (of 15)	0 exact
(5, 4)	1	1	$(\emptyset, (2, 2))$	1 (of 10)	0 exact
(7, 4)	1	3	$(\emptyset, (2, 2))$	2 (of 15)	0 exact
(9, 4)	2	1	$(\emptyset, (2, 2))$	1 (of 30)	0 exact
(11, 4)	2	3	$(\emptyset, (2, 2))$	2 (of 56)	0 exact
(9, 5)	1	4	$((5), \emptyset)$	20 (of 30)	0 exact
(13, 5)	2	3	$((5), \emptyset)$	10 (of 101)	0 exact
(7, 6)	1	1	$(\emptyset, (3, 3))$	1 (of 15)	0 exact
(9, 6)	1	3	$(\emptyset, (3, 3))$	1 (of 30)	0 exact

(Full table of all 24 pairs at `~/projects/probes/2026-08-08-type-b-self-similarity/output.txt`.)

7 Status and Next Steps

- **Empirical status:** Theorem ?? is verified with *exact* arithmetic on 24 test pairs across five values of $e \in \{2, 3, 4, 5, 6\}$ and n up to 13. The residuals are exactly zero, not merely numerical zeros.
- **Odd- e proof status:** Sketched in §3 as a mechanical translation of [?]. Full derivation of the byproduct formula (??) and one-line collapse expected to fit in ≤ 4 pages. *Marked for the next PROVE session.*
- **Even- e proof status:** Requires a strictly stronger support theorem (Observation ??) plus a companion cyclotomic identity (the $2e^2$ calculation in §4). Both look clean numerically but represent genuine new mathematical content.

References

- [1] Clio, *Self-similarity of the fake-degree Schur sum at a root of unity*, `proofs/2026-08-07-self-similarity-qe.tex`.
- [2] Clio, *Support of the fake-degree Schur sum at a root of unity*, `proofs/2026-08-06-support-conjecture-qe.tex`.
- [3] A. Ayer and D. Kumari, *Factorisations of classical group characters at odd powers of even roots of unity*, arXiv:2501.00275.
- [4] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Univ. Press, 1990.